

Create an AI Literacy Roadmap to Drive Responsible and Productive AI

1 October 2024 - ID G00821222 - 23 min read

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Initiatives: Strategy, Risks and Opportunities; People

Many organizations struggle to scale AI, with a lack of talent being one of the key reasons. The effective and responsible application of AI requires knowledge and skills throughout the organization. AI leaders should proactively design and execute an AI literacy program.

More on This Topic

This is part of 3 in-depth collections of research. See the collections:

- [Scenario Guide to AI Essentials for Digital Leaders](#)
- [Top Insights on AI for CHROs](#)
- [AI Leaders: Maturity Guide for AI Strategy](#)

Overview

Key Findings

- Generative AI has greatly accelerated the adoption of AI and its potential business impact, underlining the importance of upskilling business stakeholders on AI. Creating value from AI depends on this upskilling.
- Emerging AI literacy programs in organizations often suffer from a lack of clarity on how AI learning may lead to certain business outcomes or are overly focused on technical skills. However, productivity gains, for example, do not only require effective use of AI tools and applications but also skills to change work habits to improve AI's impact.
- A lack of talent and skills is the second top barrier to the implementation of AI techniques. Such implementations are typically multidisciplinary and require skills tailored to the needs of all those involved in both technical and nontechnical roles.
- AI technology, adoption and use are evolving rapidly with no slowdown in sight, so AI literacy skills development must match this accelerating pace of change. The 2024 Gartner Strategic Skills Development Survey found enterprises achieving the best learning outcomes at scale develop skills differently; they use agile learning practices at least 1.5 times more than others.
- AI literacy programs are not one-off, big bang exercises. Trying to educate everyone in the entire organization in one iteration is hardly feasible nor recommended, given the fact that the speed and order of AI adoption varies throughout the organization.

Recommendations

- Build the case for AI literacy and gain buy-in for investment in AI literacy by contextualizing the key benefits of AI literacy, such as the effective, productive, safe and responsible use of AI. Communicate these benefits with key stakeholders.
- Develop tangible AI literacy value propositions for each organizational unit by connecting the specific learning activities needed to realize targeted AI-enabled business outcomes.
- Identify AI literacy needs by identifying the different roles in persona groups of those involved in AI and by formulating specific objective and skills priorities for each of them, aligned with other preexisting data or digital literacy programs.
- Design and deliver an AI literacy skills development program guided by outcome-driven agile learning leading practices. Start with focused use cases and personas that can best realize business outcomes then continuously iterate and expand across the enterprise.
- Execute and monitor the delivery of the AI literacy program. Its planning should align with AI strategy priorities and AI adoption phases throughout the organization. Typically, the program should be periodically iterated, adapted and extended.

Strategic Planning Assumptions

- By 2027, 80% of large enterprises that outperform their competition will have built outcome-driven agile learning capabilities that provide the needed skills.
- By 2026, the ability to deliver data and AI literacy, culture change and a skilled workforce will be a top three determining factor in supporting business strategy.
- By 2027, more than half of organizations will secure funding for data literacy and AI literacy programs, fueled by enterprise failure to realize expected value from generative AI.

Introduction

Before organizations can reap the benefits of AI, they must first learn how to use it effectively and responsibly, with all its possibilities and limitations. However, when implementing AI, a lack of AI talent and skills is the second top barrier reported by respondents to the 2023 Gartner AI in the Enterprise Survey. ¹ Developing AI literacy, or the knowledge and skills to work with AI, throughout the organization is a critical success factor for scaling AI. Organizations not addressing AI literacy will not succeed in creating value with AI.

AI Literacy Definition

AI literacy is the ability of people to effectively and responsibly utilize AI in context (business and societal). This includes knowledge about the implications, risks and resulting value and outcomes. It includes understanding the fundamental principles of AI, technology and applications, analytical and algorithmic methods, data and knowledge sources, and ethical considerations. AI literacy, and the required depth of knowledge and competencies across these topics, vary per role of those involved in AI.

Despite advances in artificial intelligence, human intelligence remains critical. In the future, the most successful organizations will emphasize the effective synergy between humans and machines. Or, in other words, in the emerging era of AI, the importance of human talent, knowledge and learning is more important than ever before.

There is no artificial intelligence without human intelligence.

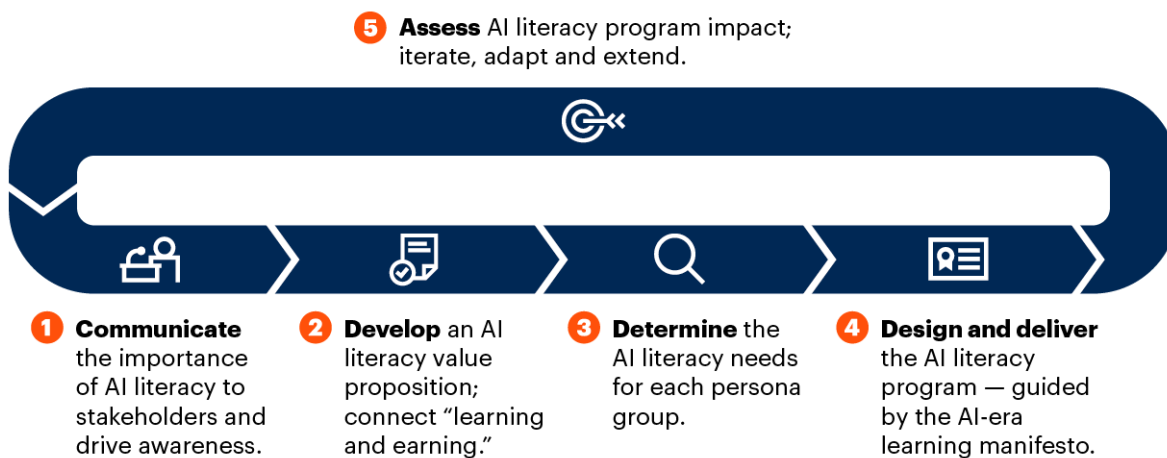
However, setting up an AI literacy program is often challenging, as the level and speed of adoption of AI varies across the organization with many stakeholders involved, each having different needs in terms of AI knowledge and skills.

Although a myriad of university and other education programs increasingly address AI knowledge and skills, many organizations seek to accelerate the maturing of AI literacy among existing and new employees. A roadmap to proactively initiate, design, execute and monitor an AI literacy program requires an iterative series of activities, organized in the five steps shown in Figure 1:

- Communicate the importance of AI literacy.
- Develop an AI literacy value proposition.
- Determine AI literacy needs for each persona group.
- Design and deliver the AI literacy program.
- Assess AI literacy program impact

Figure 1: Roadmap for an AI Literacy Program

Roadmap for an AI Literacy Program



Source: Gartner
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Gartner

Note: Any AI literacy roadmap should be aligned and coordinated with other complementary literacy and skills areas, in particular data literacy (see [CDAO Field Guide for Data Literacy and Data-Driven Business Transformation](#) and [Quick Answer: How Are AI Literacy and Data Literacy Related?](#))

Analysis

Step 1: Communicate the Importance of an AI Literacy Program to Stakeholders and Drive Awareness

To start an AI literacy program, you must build the case for its importance. This case can help obtain a mandate for an enterprisewide AI literacy program. To gain buy-in for investment in AI literacy, identify, contextualize and communicate key benefits with stakeholders. For example, by presenting AI literacy benefits — focused on those that are most relevant to the organization — to the board of directors and asking for approval to start the AI literacy program, including funding and staffing.

With AI literacy being closely related to, and overlapping with data literacy, benefits can be extrapolated from known financial performance improvements (see [Improve Your Data Literacy to Boost Financial Performance](#)). In addition, Figure 2 provides an overview of AI literacy’s key qualitative benefits. In short, AI literacy is critical to more:

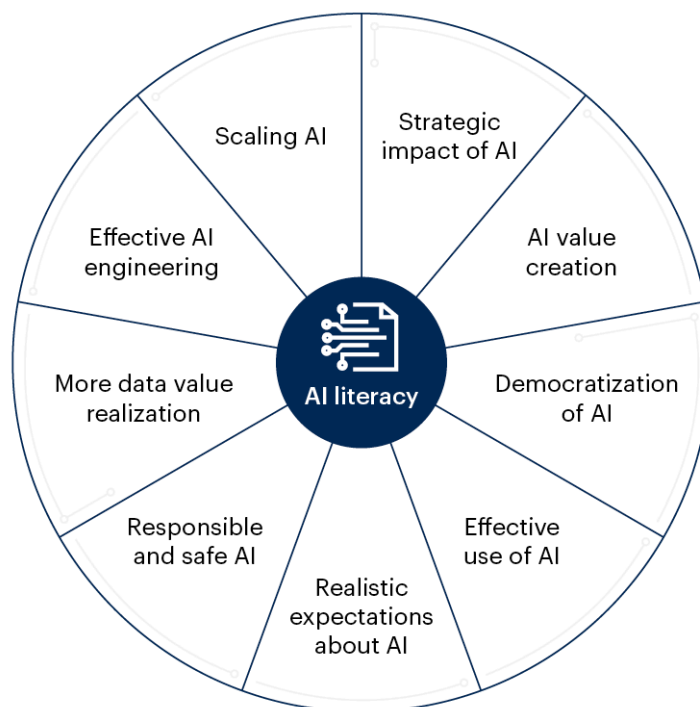
- AI scaling
- Strategic impact of AI

- AI value creation
- Democratization of AI
- Effective use of AI
- Realistic expectations about AI
- Effective AI engineering
- Data value realization
- Responsible and safe AI

Please see Note 1 for further descriptions of these benefits.

Figure 2: Key Benefits of AI Literacy

Key Benefits of AI Literacy



Source: Gartner
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Step 2: Develop an AI Literacy Value Proposition by Connecting “Learning to Earning”

An enterprise literacy program’s overall value proposition is to enable the organization to execute on its AI ambition and AI strategy by maximizing the realization of results (see [The Pillars of a Successful Artificial Intelligence Strategy](#) and [Gartner AI Opportunity Radar: Set Your Enterprise’s AI Ambition](#)).

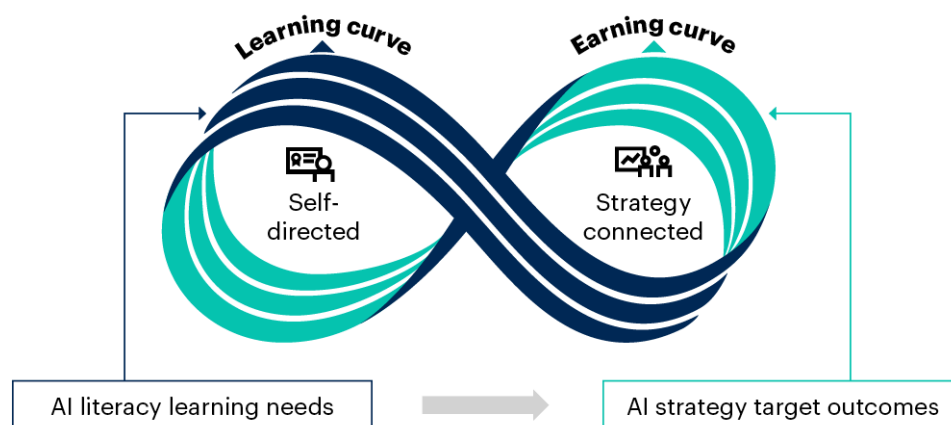
Is AI only to be used in some narrow “everyday” use cases, or is AI seen as a strategically important gamechanger? If the former, AI literacy can be limited to specific employee groups at a relatively basic level. If the latter, then AI literacy is critical and should be developed proactively and pervasively throughout the enterprise.

To ensure the AI literacy program is not too widely scoped and instead focuses on the highest value and most realizable outcomes within the AI strategy and ambition, AI leaders should link “learning to earning.” They should create narratives with their teams that connect the specific learning activities needed to realize targeted AI-enabled business outcomes (see Figure 3).

Figure 3: Co-Created “Learning to Earning” Narratives

Co-Created “Learning to Earning” Narratives

Illustrative



Source: Gartner
821222_C

Gartner

Each “learning to earning” narrative identifies specific AI literacy needs for specific employee personas to fulfill prioritized AI use cases. Stakeholders include business managers, HR learning and development teams, talent and organization change teams, affected employees, and, particularly, the executive or leader with accountability for the business outcome. This connection between the learning activities and achieving business outcomes should yield the primary success metrics for the AI literacy program. See Table 1 for examples.

Table 1: Example ‘Learning to Earning’ Value Propositions for AI Literacy

AI Literacy “Learning” (Needs)	AI Strategy “Earning” (Target Outcomes)
Upskill marketing teams with data and AI literacy skills to harness domain-specific vendor AI marketing solutions.	Improve marketing campaign performance by 20% via greater AI-enabled personalization and targeting.
Equip software engineering employees with skills to harness vendor provided AI-augmented software development tools.	Boost productivity of software development by 5% while delivering products with at least 20% better quality metrics.
Build skills in the core AI team to create a domain-specific model and AI literacy skills in pharma R&D scientists to use it.	Discover new molecules and medicines 50% faster that gain regulatory approval with 10% higher probability.
Develop AI literacy across the entire workforce to leverage everyday AI tools and execute a game changing AI strategy.	Increase overall employee productivity by 10% and leverage differentiating AI capabilities to gain 30% market share.

Source: Gartner

Step 3: Determine AI Literacy Needs for Each Persona Group

Identify Persona Groups

Regardless of ambition levels, the effective and responsible use of AI requires knowledge and skills for both technical and business/nontechnical roles — anyone involved in the prioritization, ideation, development, use and operations of AI applications. These roles include but are not limited to:

- Executives who need to decide on AI investments

- Business subject matter experts (SMEs) who need to be part of implementation projects, supporting design decisions, testing features, ensuring business relevance and outcomes
- AI experts who need to design, build, deploy and support AI solutions
- Employees who need to use the AI capabilities provided

Different people have different AI literacy needs. A one-size-fits-all AI literacy program does not exist.

AI literacy needs should be identified for each target audience or persona group, as far as they relate to the value propositions as identified in Step 2. See Table 2 for examples.

Table 2: Examples of AI Literacy Needs by Persona Group
(Enlarged table in Appendix)

Persona Group	AI Literacy Needs
Business Leaders (C-level executives, business management, innovation and change managers)	- Ability to assess the current and future strategic impact of AI - The knowledge to set strategic priorities for AI goals aligned with business strategy, competitive position and market conditions - Skills to make more informed AI investment decisions and drive and own AI literacy effectively - Knowledge about AI's strengths, risks and limitations to identify and prioritize AI use cases - Understanding of how AI can be a catalyst for both incremental and radical innovation - Knowledge about AI to assess its impact on existing or new processes, products or services and make changes accordingly
Business Operations and Execution (AI users - including internal employees, business SMEs, data and AI domain experts, frontlines business analysts)	- Know how to use AI effectively to work and how to change work habits - Awareness about AI risks and limitations and understand policies and guidelines - Knowledge about AI's strengths and limitations to identify and prioritize AI use cases - Effective collaboration with AI experts and engineers to implement effective AI solutions
Business Staff (HR partners, finance, legal, compliance and audit, corporate governance, partner managers)	- Skills to capture and monitor AI business impact, value and costs, while mitigating risks - Knowledge about AI, regulations, risks and obligations - Effective involvement in AI governance, in collaboration with AI leaders and other stakeholders - Understanding of AI to assess the impact of AI on job profiles, education and skills, work conditions, and employability - The ability to identify new AI-related roles or changes to existing ones for effective recruitment, training or upskilling
AI Leaders and Experts (AI leaders, AI architects, AI governance data scientists, AI experts, analytics experts, domain engineers, knowledge engineers, prompt engineers)	- Knowledge about current and emerging AI - Ability to identify and assess AI capabilities, in particular AI team roles and skills required - Skills to identify and enhance the evolution of the AI strategy - Knowledge and skills to identify design, build or outsource, deploy and monitor AI solutions - Deep knowledge about a growing number of AI approaches, techniques, models, tools and frameworks - Effective collaboration with business domain experts, data and software engineers and other stakeholders - Knowledge about AI applications across such as data science, decision intelligence, virtual assistants or smart vehicles
Engineers and Operators (Data managers and engineers, software engineers, model validators, software quality integrators, IT operators, infrastructure specialists)	- Understanding of AI and techniques, including their applicability as well as how to combine and integrate them with their software and systems - Understanding of AI requirements with respect to data acquisition, integration and preparation of real-world data - Effective collaboration with AI experts and engineers to implement AI solutions - Knowledge to register and manage AI (digital) infrastructure, platforms, data, tools and applications and monitoring of AI - Knowledge for operations with deployment, operations and monitoring of AI - Skills for providing first-line user support and dealing with AI production issues
Data and Technology Leaders (IT management, enterprise architects, IT architects, data and analytics leaders, data governance, data architects)	- Knowledge about current and emerging AI - Understanding of how AI relates to other (digital) technology - Understanding of AI requirements with respect to data governance and data management

Source: Gartner

Determine AI Literacy Levels for Each Persona Group

In collaboration with HR, assess current AI literacy levels to design a training program that closes each persona group’s skills gaps. The AI literacy program should align with and complement other preexisting data or digital literacy programs (see [Quick Answer: How Are AI Literacy and Data Literacy Related?](#) and [How to Lead Upskilling Initiatives for Data Scientists and AI Experts](#))..

Any training curriculum for AI literacy programs consists of four main categories about AI foundations, value, engineering and governance (see Table 3).

Table 3: AI Literacy Categories

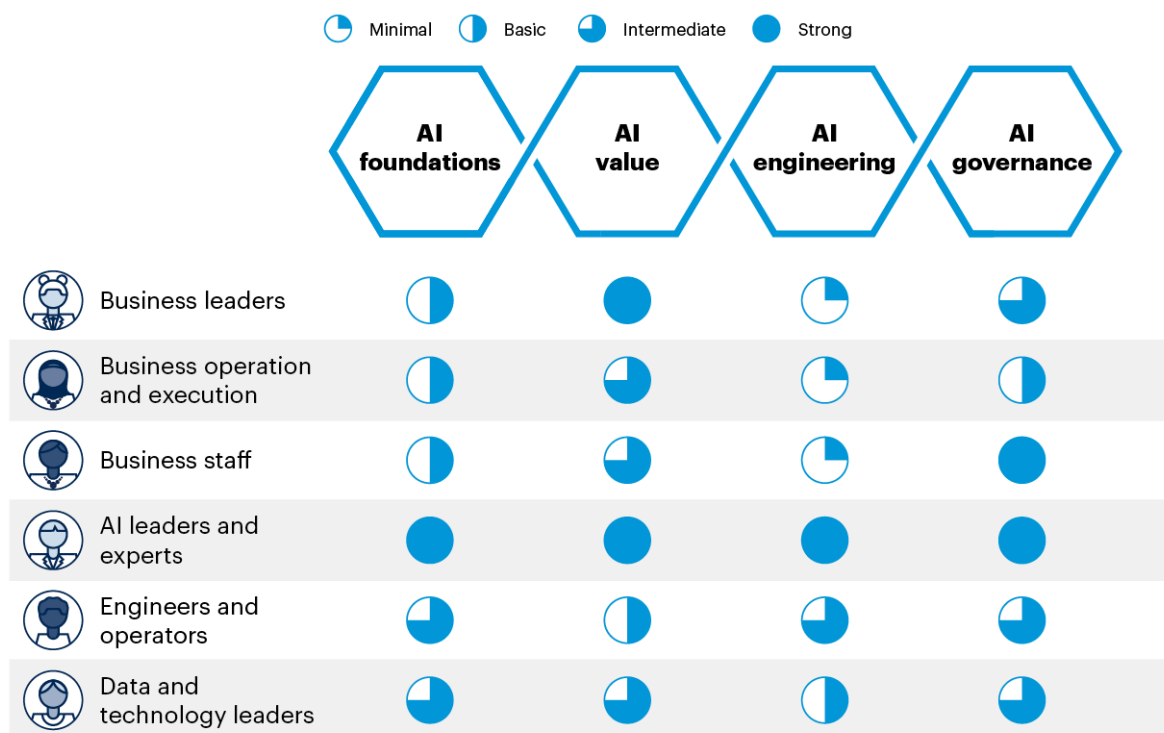
(Enlarged table in Appendix)

AI Literacy Category ↓	Elements ↓
AI Foundations	■ Key concepts (AI, machine learning, deep learning, data science, symbolic AI, generative AI) ■ Main application areas (decision intelligence, predictive analytics and data science, virtual assistants, smart robotics, etc.) ■ Main techniques, methods, practices
AI Value	■ Use cases, benefits and costs ■ Business domain expertise ■ Effective use, evaluation ■ Change management, culture
AI Engineering	■ Design, buy-build, data preparation, model selection/fine-tuning/training, validation, deployment, monitoring, operations ■ Algorithm and model types (data-driven such as foundation models and symbols such as knowledge graphs) ■ Data management and engineering, analytics ■ Other (complementary) approaches and techniques (AI agents, composite AI, optimization, etc.)
AI Governance	■ Regulations, policies, ethics, societal aspects ■ Risk (reliability/accuracy, security, safety, fairness, privacy, intellectual property, etc.) assessment and mitigation ■ Trust, transparency, explainability (systems, models, decisions) ■ Data and other governance

Source: Gartner

Not all persona groups need to have knowledge about these categories at the same level of depth. See the persona groups in Figure 4. The figure indicates each persona group's the required skill levels for the four main categories in an AI literacy training curriculum

Figure 4: Persona Groups and Required AI Literacy Skill Levels

Persona Groups and Required AI Literacy Skills Levels

Source: Gartner
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Gartner

AI literacy needs and required skill levels form the foundation for more detailed skills profiles per role (see [Core Generative AI Skills Readiness Assessment for a Diverse AI Team](#) and [Quick Answer: What Is an AI Leader?](#)), as well as the design and delivery of an AI literacy program.

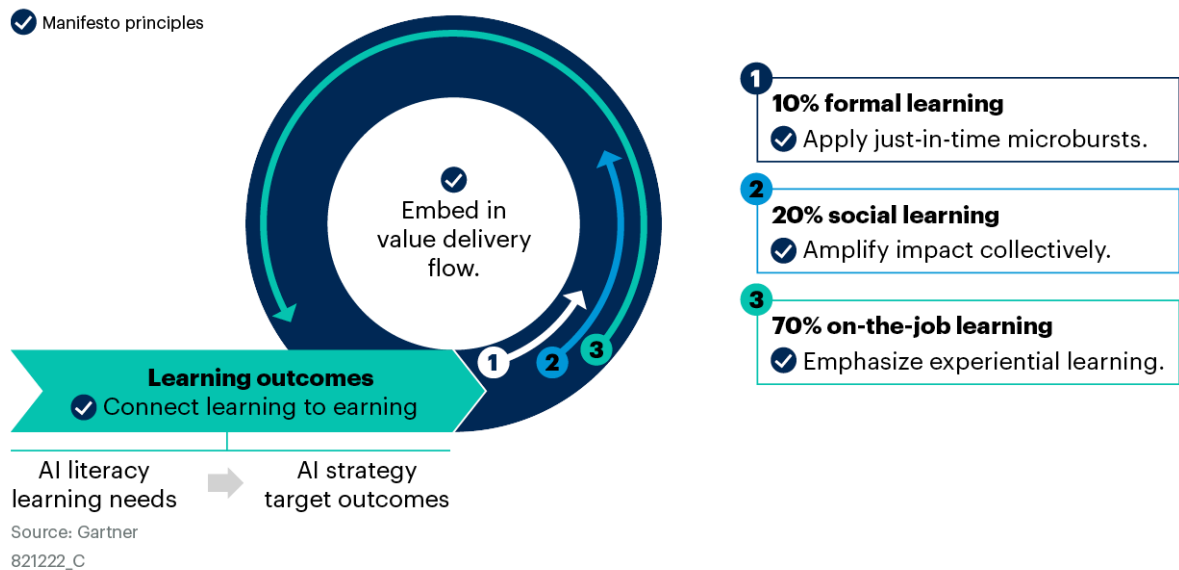
Step 4: Design and Deliver the AI Literacy Program Guided by the AI-Era Learning Manifesto

AI technology, adoption, use, capabilities and associated risks are evolving rapidly with no slowdown in sight, so AI literacy skills development must match this accelerating pace of change. It must also span from relatively basic skills for end users of provisioned AI applications to deep expert proficiency, such as training machine learning models or other advanced AI capabilities. Lastly, it must be able to scale quickly from early adopters to the entire enterprise workforce. AI leaders should therefore adopt outcome-driven agile learning practices proven to develop skills rapidly, scale across an entire workforce and enable strategic business outcomes (see [AI-Era Learning Manifesto: Outcome-Driven Agile Learning](#)).

AI leaders should start with the “learning to earning” narratives from Step 2 that define the highest priority AI literacy needs to guide the program design. Then, adopt the AI-era learning manifesto principle of “embedding learning in the value delivery flow” (see Figure 5), a practice that is two times more likely in enterprises that deliver desired outcomes from learning, according to the 2024 Gartner Strategic Skills Survey. ²

Figure 5: Embedded Learning in Value Delivery Activities

Embedded Learning in Value Delivery Activities



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Leaders should ensure learning pathways combine formal learning, social learning and on-the-job application of new skills via experiential learning to achieve the desired business outcome. The ratio will vary depending on the specific AI literacy skills, level of proficiency needed and outcomes targeted, but a good rule of thumb is 10-20-70 (see Figure 5 above).

Formal Learning

Leaders can partner with L&D teams and domain experts to curate learning curriculum courses from AI technology partners, AI learning service providers and educational institutions — often via learning platforms. These courses should be augmented by learning resources, such as short educational videos, created by early adopter practitioners and learning champions. Where possible, the formal learning should follow the AI-era learning manifesto principle of “apply just-in-time microbursts.”

Apply just-in-time microbursts.

People learn best in short, well-designed chunks, applied right away, then practiced regularly. These chunks can be combined to deepen proficiency and develop skills incrementally at pace. Bite-size learning experiences can be consumed anytime, anywhere and brought to the learner at the moment of need. Microbursts underpin continuous agile learning.

Social Learning

The AI literacy program should utilize communities of practice and centers of excellence to bring people together for knowledge sharing and collective learning. These social learning groups should exist at the persona level, focused on AI skills applied to similar roles, and across departments and the enterprise to stimulate and spread learning. Activities include collective learning events, hands-on group exercises, sharing prompt libraries and gamification. Harness the AI-era learning manifesto principle of “amplify impact collectively” to guide the social learning.

Amplify impact collectively.

Social connections power people to help each other learn. Collective learning cohorts and communities of practice spread new skills and confidence across teams. A vibrant continuous learning culture amplifies the impact of new skills and capabilities via network effects. Community and culture amplify how learning drives outcomes at scale.

On-the-Job Experiential Learning

For the new AI literacy skills to impact business outcomes, learners must apply new AI literacy skills from the formal learning on-the-job. This phase should be supported by the social learning structures including coaching, mentoring, shadowing, paired application of skills and vendor knowledge transfer. Other methods include “learning by doing” via experiments, proof of concepts and hackathons (see [Improve AI Literacy and Skills by Running an AI Hackathon](#)). The AI-era learning manifesto principle “emphasize experiential learning” shows the importance of this element in the learning pathway.

Emphasize experiential learning.

Learning by doing — particularly on the job (or in personal life) in the flow of achieving an outcome — ensures new skills are really learned and readily retained and result in more value. While learning pathways include new theory and support such as coaching, up to 70% of what makes learning effective is experiential, such as hands-on challenging experiences and assignments.

AI leaders should rightsize facets of an overall AI literacy program based on the employee persona groups, proficiency levels and outcomes targeted. This can range from relatively low proficiency and fast skills development for end users of enterprise-provisioned and data secured AI applications to expert proficiency via extensive skills development data and AI literacy personas building custom-made machine learning models or fine-tuning foundation models, for example.

Step 5: Assess AI Literacy Program Impact — Iterate, Adapt and Extend

AI leaders should regularly (i.e., quarterly) conduct a review or retrospective to assess the four previous steps of the AI literacy roadmap. This review should explore whether the program is developing the needed AI literacy to enable business outcomes. This should be combined with a review of data literacy activities, if existing (see [3 Ways to Measure and Communicate the Value of Data Literacy](#)).

The program should also be agile in design with ability to rapidly respond to changing events such as competitors' AI strategies, business leaders' heightened interest and/or new compelling business cases, emerging or undermitigated risks, technical developments, and new AI capabilities. Key considerations for each step include:

- Step 1 — Evolving stakeholder appetite and AI ambition in context of overall business strategy or government mission, AI's ability to contribute to core business performance metrics and competitive posture.
- Step 2 — Emerging new high potential use cases with high value creation or existing use cases not living up to the promise or exposing the organization to unacceptable risks (This should include ethical risks and new or updated AI regulations.).
- Step 3 — Changing AI literacy needs throughout the entire enterprise with emphasis where skills gaps are either impeding AI initiative execution or exposing the organization to risks.

- Step 4 – Efficacy of the learning program design and delivery:
 - Is the needed formal training available, sufficiently tailored for use cases and personas and rated highly by employees?
 - Are there effective social learning practices such as communities of practices and coaching to enable collective learning and teaching?
 - Most importantly, are employees able to apply the new AI literacy skills in their jobs and contribute to achieving targeted outcomes?

Based on the review, the AI literacy program should adapt to keep pace with the enterprise's AI ambition and strategy while building an AI-ready workforce adept in harnessing AI to drive targeted outcomes.

Evidence

¹ **2023 Gartner AI in the Enterprise Survey.** This study was conducted to understand the keys to successful AI implementations and their impact on the broader AI that has been brought by generative AI. The research was conducted online from 19 October through 21 December 2023 among 703 respondents from organizations in the U.S., Germany and the U.K. The main sample consisted of 645 out of the 703. Organizations were required to have developed or intended to deploy at least two AI initiatives within the next three years. Respondents were required to be part of the organization's corporate leadership or report to corporate leadership roles. Fifty-eight out of 703 are the business intelligence (BI) sample. Organizations were required to have developed or intended to deploy at least one AI initiative within the next three years. Respondents were required to be part of the organization's corporate leadership or report to corporate leadership roles or below (senior manager and above) and to be primarily responsible for BI in their organizations. Both the main sample and the BI sample respondents were required to have a high level of involvement with at least one AI initiative, and they were required to have one of the following roles when related to AI in their organizations: determine AI business objectives, measure the value derived from AI initiatives, or manage AI initiatives development and implementation. Quotas among the main sample were established for company size and for industries to ensure a good representation across the sample. No quotas were established for the BI sample. Disclaimer: The results of this survey do not represent global findings or the market as a whole, but reflect the sentiments of the respondents and companies surveyed.

² **2024 Gartner Strategic Skills Development Survey.** This survey was conducted to understand the current status of agile learning, specifically with an eye to agile learning execution and the ways in which such learning supports key C-level priorities. The research was conducted online from 27 April through 13 June 2024 among 330 respondents from North America (n = 123), EMEA (n = 115) and Asia/Pacific (n = 92), with revenue distributed across midsize (n = 103), large (n = 107) and global enterprises (n = 120). Respondents were required to be involved in training and learning at their organization. Disclaimer: Results of this study do not represent global findings or the market as a whole but reflect sentiment of the respondents and companies surveyed.

Note 1: Key Benefits of AI Literacy

AI Literacy is critical to more:

- **Scaling AI:** To move beyond AI experimentation and piloting and to apply AI more broadly and systemically, a general boost of knowledge and awareness across the organization is needed. Moreover, the strategic and pervasive use of AI requires an AI-first culture and mindset, fostered by literacy across the organization.
- **Strategic impact of AI:** Market conditions or competitive pressure may require to go beyond the use of everyday AI, toward more game-changing AI (see [Gartner AI Opportunity Radar: Set Your Enterprise's AI Ambition](#)). However, realizing the potential of AI as a competitive differentiator, requires literacy up to the most senior levels of leadership. This greatly helps to align the more innovative use of AI with strategic business priorities, and vice versa.
- **AI value creation:** The ideation, selection and prioritization of AI use cases is easier said than done. Business stakeholders need to understand the key strengths and weaknesses of AI to align its use with key business goals, challenges and opportunities. AI literacy among all stakeholders reduces the noise in communication between technical and nontechnical people while exploring AI application possibilities.
- **Support democratization of AI:** In recent years, AI has become relatively easier to apply, bringing it within reach of many people and organizations. This has been made possible through further automation, greater user friendliness and the growing availability of foundation models. Nonetheless, the effective and responsible use of AI still requires a level of skills that depends on the context and complexity of the AI being used. Or, in other words: Democratization should go hand in hand with education.

- **Effective use of AI:** Individuals who are using AI, require knowledge and skills on how to process input for AI (e.g., prompting in the case of chatbot-based generative AI) and output of AI (e.g., how to interpret a prediction of a data science model). Moreover, skills are needed to incorporate the use of AI in one's way of working, changing existing work habits to maximize AI's impact in terms of gains in productivity or quality. In practice, without such skills, AI's value creation is often much smaller (see [Who Benefits Most From Generative AI Productivity?](#)).
 - **Realistic expectations about AI:** Due to the enormous hype around AI, people can be overly optimistic about AI as a miracle cure, or overly pessimistic about AI as a threat to employment and humanity at large. Both extremes often result from limited knowledge about what current day AI is, and what it is not. AI literacy is critical to bring positive and negative sentiments back to realistic proportions. This is essential for successful use of AI, as it helps to avoid disappointment about AI's limitations and to reduce resistance to AI initiatives.
 - **Effective AI engineering:** The design, development and operation of AI solutions require technical knowledge and skills. This is also true for AI that is supplied by third parties (vendors, partners or open source), as the selection, acquisition and the – often much needed – contextualization or customization of AI also require literacy. Such literacy is not only needed by AI experts but also applies to other roles involved in AI engineering, such as data engineers and software engineers.
 - **More data value realization:** In addition to investments in data and analytics, AI can help to further turn the value potential of data into reality. Analytics complemented with data science and generative AI can boost data-driven insights and better decision intelligence. This, however, requires multidisciplinary collaboration between data and analytics, AI and business professionals for which both data and AI literacy are indispensable.
 - **Responsible and safe AI:** Risks regarding ethics, reliability, intellectual property and security cannot be managed by policies, regulatory compliance or tools alone. Effective AI governance for safe and responsible AI benefits from awareness and sufficient knowledge and skills among all relevant stakeholders. This should not be limited to legal or compliance roles but also applies to business stakeholders and people involved in the engineering and use of AI.
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Recommended by the Authors

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[Without AI Literacy, ROI Plummets](#)

Improve AI Literacy and Skills by Running an AI Hackathon

Core Generative AI Skills Readiness Assessment for a Diverse AI Team

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Table 1: Example ‘Learning to Earning’ Value Propositions for AI Literacy

AI Literacy “Learning” (Needs)	AI Strategy “Earning” (Target Outcomes)
Upskill marketing teams with data and AI literacy skills to harness domain-specific vendor AI marketing solutions.	Improve marketing campaign performance by 20% via greater AI-enabled personalization and targeting.
Equip software engineering employees with skills to harness vendor provided AI-augmented software development tools.	Boost productivity of software development by 5% while delivering products with at least 20% better quality metrics.
Build skills in the core AI team to create a domain-specific model and AI literacy skills in pharma R&D scientists to use it.	Discover new molecules and medicines 50% faster that gain regulatory approval with 10% higher probability.
Develop AI literacy across the entire workforce to leverage everyday AI tools and execute a game changing AI strategy.	Increase overall employee productivity by 10% and leverage differentiating AI capabilities to gain 30% market share.

Source: Gartner

Table 2: Examples of AI Literacy Needs by Persona Group

Persona Group	AI Literacy Needs
Business Leaders (C-level executives, business management, innovators and change managers)	■ Ability to assess the current and future strategic impact of AI ■ The knowledge to set strategic yet realistic AI goals, aligned with business strategy, competitive pressure and market conditions ■ Skills to make more informed AI investment decisions and steer and own AI initiatives effectively ■ Knowledge about AI's strengths, risks and limitations to identify and prioritize AI use cases ■ Understanding on how AI can be a catalyst for both incremental and radical innovation ■ Knowledge about AI to assess its impact on existing or new processes, products or services and make changes accordingly
Business Operations and Execution (AI users — including internal employees, business SMEs, data and AI domain stewards, translators, business analysts)	■ Know how to use AI effectively in work and how to change work habits Awareness about AI risks and limitations and understand policies and guidelines ■ Knowledge about AI's strengths and limitations to identify and prioritize AI use cases

- Effective collaboration with AI experts and engineers to implement effective AI solutions

Business Staff (HR partners, finance, legal, compliance and audit, corporate governance, partner managers)

- Skills to estimate and monitor AI business impact, value and costs, while mitigating risks
- Knowledge about AI, regulations, risks and mitigations
- Effective involvement in AI governance, in collaboration with AI leaders and other stakeholders
- Understanding of AI to assess the impact of AI on job profiles, education and skills, work conditions, and employability
- The ability to identify new AI-related roles or changes to existing roles for effective recruitment, reskilling or upskilling

AI Leaders and Experts (AI leader, AI architects, AI governance data scientists, AI experts, analytics experts, decision engineers, knowledge engineers, prompt engineers)

- Knowledge about current and emerging AI
- Ability to identify and mature AI capabilities, in particular AI team roles and skills required
- Skills to develop and orchestrate the execution of the AI strategy
- Knowledge and skills to ideate, design, build or customize, deploy, and monitor AI solutions
- Deep knowledge about a growing number of AI approaches, techniques, models, tools and frameworks

- Effective collaboration with business domain experts, data and software engineers and other stakeholders
- Knowledge about AI application areas such as data science, decision intelligence, virtual assistants or smart robotics

Engineers and Operators

(Data managers and engineers, software engineers, model validators, software testers, integrators, IT operations, infrastructure specialists)

- Understanding of AI and AI techniques, including their applicability as well as how to combine and integrate them with other software and systems
- Understanding of AI requirements with respect to data acquisition, integration and preparation of multistructured data
- Effective collaboration with AI experts and engineers to implement AI solutions
- Knowledge to explore and manage AI (cloud) infrastructure, platforms, data, tools and applications
- Knowledge for (assistance with) deployment, operations and monitoring of AI
- Skills for providing first line end-user support and dealing with AI production issues

Data and Technology Leaders (IT management, enterprise architects, IT architects, data and analytics leaders, data governance, data architects)

- Knowledge about current and emerging AI
- Understanding of how AI relates to other (digital) technology
- Understanding of AI requirements with respect to data governance and data management

Source: Gartner

Table 3: AI Literacy Categories

AI Literacy Category ↓	Elements ↓
AI Foundations	■ Key concepts (AI, machine learning, deep learning, data science, symbolic AI, generative AI) ■ Main application areas (decision intelligence, predictive analytics and data science, virtual assistants, smart robotics, etc.) ■ Main techniques, methods, practices
AI Value	■ Use cases, benefits and costs ■ Business domain expertise ■ Effective use, evaluation ■ Change management, culture
AI Engineering	■ Design, buy-build, data preparation, model selection/fine-tuning/training, validation, deployment, monitoring, operations ■ Algorithm and model types (data-driven such as foundation models and symbols such as knowledge graphs) ■ Data management and engineering, analytics

AI Literacy Category ↓	Elements ↓
	■ Other (complementary) approaches and techniques (AI agents, composite AI, optimization, etc.)
AI Governance	■ Regulations, policies, ethics, societal aspects ■ Risk (reliability/accuracy, security, safety, fairness, privacy, intellectual property, etc.) assessment and mitigation ■ Trust, transparency, explainability (systems, models, decisions) ■ Data and other governance

Source: Gartner